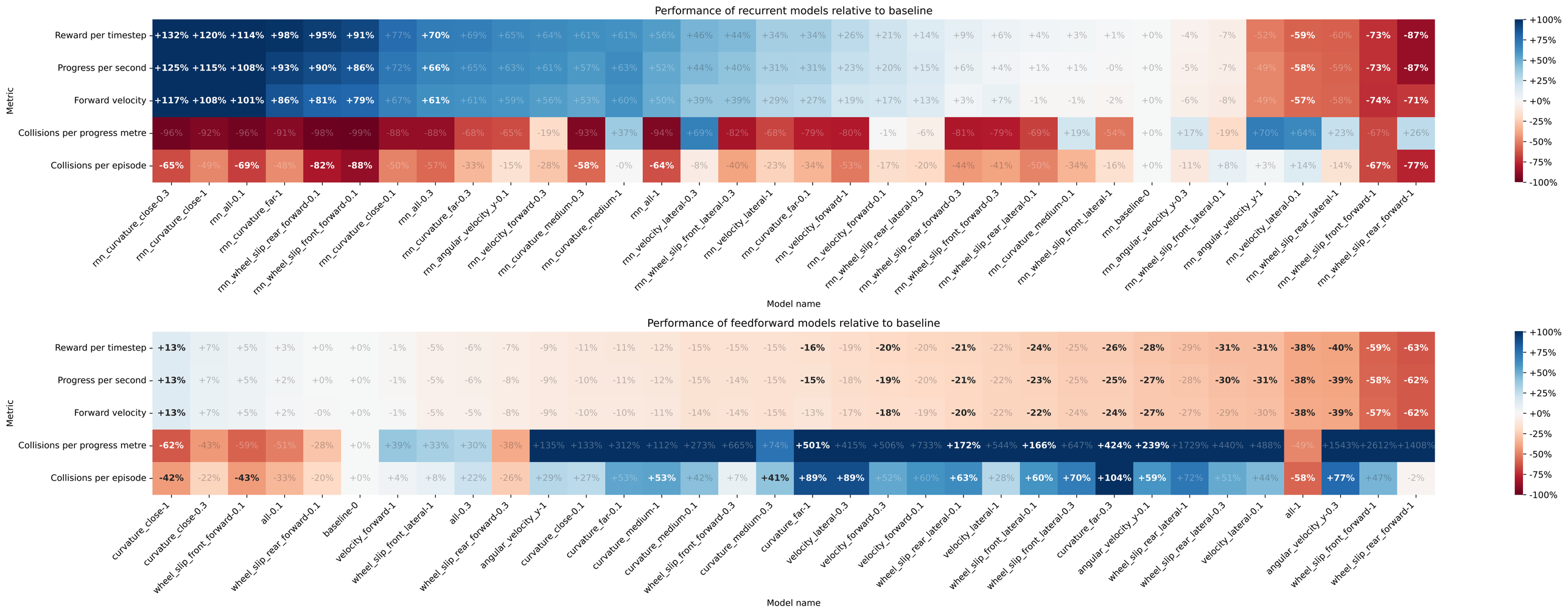
Performance of recurrent models relative to baseline
Metric
Reward per timestep
Progress per second
Forward velocity
Collisions per progress metre
Collisions per episode
+132% +120% +114% +98% +95% +91% +77% +70% +69% +65% +64% +61% +61% +56% +46% +44% +34% +34% +26% +21% +14% +9% +6% +4% +3% +1% +0% -4% -7% -52% -59% -60% -73% -87%
+125% +115% +108% +93% +90% +86% +72% +66% +65% +63% +61% +57% +63% +52% +44% +40% +31% +31% +23% +20% +15% +6% +4% +1% +1% -0% +0% -5% -7% -49% -58% -59% -73% -87%
+117% +108% +101% +86% +81% +79% +67% +61% +61% +59% +56% +53% +60% +50% +39% +39% +29% +27% +19% +17% +13% +3% +7% -1% -1% -2% +0% -6% -8% -49% -57% -58% -74% -71%
-96% -92% -96% -91% -98% -99% -88% -88% -68% -65% -19% -93% +37% -94% +69% -82% -68% -79% -80% -1% -6% -81% -79% -69% +19% -54% +0% +17% -19% +70% +64% +23% -67% +26%
-65% -49% -69% -48% -82% -88% -50% -57% -33% -15% -28% -58% -0% -64% -8% -40% -23% -34% -53% -17% -20% -44% -41% -50% -34% -16% +0% -11% +8% +3% +14% -14% -67% -77%
rnn_curvature_close-0.3
rnn_curvature_close-1
rnn_all-0.1
rnn_curvature_far-1
rnn_wheel_slip_rear_forward-0.1
rnn_wheel_slip_front_forward-0.1
rnn_curvature_close-0.1
rnn_all-0.3
rnn_curvature_far-0.3
rnn_angular_velocity_y-0.1
rnn_velocity_forward-0.3
rnn_curvature_medium-0.3
rnn_curvature_medium-1
rnn_all-1
rnn_velocity_lateral-0.3
rnn_wheel_slip_front_lateral-0.3
rnn_velocity_lateral-1
rnn_curvature_far-0.1
rnn_velocity_forward-1
rnn_velocity_forward-0.1
rnn_wheel_slip_rear_lateral-0.3
rnn_wheel_slip_rear_forward-0.3
rnn_wheel_slip_front_forward-0.3
rnn_wheel_slip_rear_lateral-0.1
rnn_curvature_medium-0.1
rnn_wheel_slip_front_lateral-1
rnn_baseline-0
rnn_angular_velocity_y-0.3
rnn_wheel_slip_front_lateral-0.1
rnn_angular_velocity_y-1
rnn_velocity_lateral-0.1
rnn_wheel_slip_rear_lateral-1
rnn_wheel_slip_front_forward-1
rnn_wheel_slip_rear_forward-1
Model name
+100%
+75%
+50%
+25%
+0%
-25%
-50%
-75%
-100%
Performance of feedforward models relative to baseline
Metric
Reward per timestep
Progress per second
Forward velocity
Collisions per progress metre
Collisions per episode
+13% +7% +5% +3% +0% +0% -1% -5% -6% -7% -9% -11% -11% -12% -15% -15% -15% -16% -19% -20% -20% -21% -22% -24% -25% -26% -28% -29% -31% -31% -38% -40% -59% -63%
+13% +7% +5% +2% +0% +0% -1% -5% -6% -8% -9% -10% -11% -12% -15% -14% -15% -15% -18% -19% -20% -21% -22% -23% -25% -25% -27% -28% -30% -31% -38% -39% -58% -62%
+13% +7% +5% +2% -0% +0% -1% -5% -5% -8% -9% -10% -10% -11% -14% -14% -15% -13% -17% -18% -19% -20% -22% -22% -24% -24% -27% -27% -29% -30% -38% -39% -57% -62%
-62% -43% -59% -51% -28% +0% +39% +33% +30% -38% +135% +133% +312% +112% +273% +665% +74% +501% +415% +506% +733% +172% +544% +166% +647% +424% +239% +1729% +440% +488% -49% +1543% +2612% +1408%
-42% -22% -43% -33% -20% +0% +4% +8% +22% -26% +29% +27% +53% +53% +42% +7% +41% +89% +89% +52% +60% +63% +28% +60% +70% +104% +59% +72% +51% +44% -58% +77% +47% -2%
curvature_close-1
curvature_close-0.3
wheel_slip_front_forward-0.1
all-0.1
wheel_slip_rear_forward-0.1
baseline-0
velocity_forward-1
wheel_slip_front_lateral-1
all-0.3
wheel_slip_rear_forward-0.3
angular_velocity_y-1
curvature_close-0.1
curvature_far-0.1
curvature_medium-1
curvature_medium-0.1
wheel_slip_front_forward-0.3
curvature_medium-0.3
curvature_far-1
velocity_lateral-0.3
velocity_forward-0.3
velocity_forward-0.1
wheel_slip_rear_lateral-0.1
velocity_lateral-1
wheel_slip_front_lateral-0.1
wheel_slip_front_lateral-0.3
curvature_far-0.3
angular_velocity_y-0.1
wheel_slip_rear_lateral-1
wheel_slip_rear_lateral-0.3
velocity_lateral-0.1
all-1
angular_velocity_y-0.3
wheel_slip_front_forward-1
wheel_slip_rear_forward-1
Model name
+100%
+75%
+50%
+25%
+0%
-25%
-50%
-75%
-100%